Welcome to SPACECODE Challenge 2026!

Hosted by the Space Technology and Astronomy Club (STAC), *IIT Mandi*

The Challenge starts at: 10th January, 00:00 Hrs
Submissions close at: 11th January, 23:59 Hrs

STAC presents to you this year's **SPACECODE** challenge! Get ready to explore **computational astronomy**, **classical machine learning**, **deep learning** and **theoretical physics** through focused problem statements based on real scientific work done across universities and space organizations. This is a **2-day hackathon** designed for teams who are new to these competitions but ready to work like engineers: build baselines early, iterate with evidence, and communicate clearly.

Team size: 2–3 participants (recommended)
Allowed tools: Python + any ML framework (PyTorch/TensorFlow/JAX), common libraries, notebooks/scripts.
Note: Mention and cite the resources and codes if you have referred them from somewhere (e.g. Kaggle, GitHub). Straight plagiarism will attract penalties.

Register your team at: `https://forms.gle/979Cw8d8WWFqsLyo9`

In addition to the computational tasks, we include a small set of theory questions. These do not need coding but can give you an edge if answered with good reasoning.

Marking Scheme: Attempt any 2 computational questions out of the given 3. Though each computational question might not have equal weightage, participants need to keep in mind the total score will be based on the final results, code completion and quality.

- **Problem-1: 30 points**
- **Problem-2: 40 points**
- **Problem-3: 30 points**

You also have to attempt the theoretical questions with handwritten solutions instead of word files or latex documentation. Your approach will determine your final score. **In theoretical, each question has equal weightage. Total theory points = 40.**

Computational Questions

0.1 Problem Statement: The Galaxy Morphology Classification Challenge

1. Background: Galaxy Morphology

Galaxy morphology (spiral, edge-on, merging, etc.) is a useful proxy for a galaxy's formation history. Astronomers must distinguish between these visual structures to understand how different stellar systems evolved over billions of years. However, identifying specific morphology is complex. Distinguishing features like a "Bar" in a spiral galaxy or identifying merging systems requires high-resolution analysis of structural traits to confirm a classification.

2. Objective

The objective of this task is to build a **multi-class classification model** that predicts the **Morphology Class** of a galaxy from an input image. Using planetary and stellar metadata (in the form of image pixels), your model must distinguish between 10 categories, including:

- **Spiral & Barred Spiral**: Galaxies with distinct arm or central bar structures.
- **Merging**: Systems in the process of gravitational interaction or collision.
- **Edge-on with Bulge**: Disk galaxies viewed from a lateral perspective.

3. Data Description

The dataset contains physical image parameters derived from astronomical observations. The features and splits are grouped as follows:

- **Image Data**: 256×256 RGB galaxy images.
- **Label Field**: A **target** field representing 10 morphological categories.
- **Data Splits**: A training set of 14,188 samples and a validation set of 3,548 samples.

4. Task 01: Core Classification Tasks

- **Data Exploration**: Verify the label mapping directly from the dataset features to identify the 10 morphology classes.
- **Preprocessing**: Build a working training and evaluation baseline pipeline to establish initial performance.
- **Model Training**: Improve the baseline using a principled approach, such as better architecture, augmentation, or tuning.
- **Evaluation**: Measure performance using a confusion matrix and per-class metrics like precision, recall, and F1-score.
- **Interpretation**: Identify which classes are most confused with each other and discuss the visual reasons for these errors.

5. Task 02: Generalization & Robustness (Mini-Extension)

A model that only works on "clean" images is fragile. Your goal here is to test robustness with small, realistic perturbations. Choose one track to investigate:

- **Track 1: Augmentation study**: Pick 3 augmentations you believe are valid (do not change the class meaning) and quantify impact.
- **Track 2: Noise/blur test**: Evaluate your best model under mild blur, noise, or compression and report how performance changes.
- **Track 3: Calibration**: Check if your predicted probabilities are overconfident using a simple reliability plot or ECE.

6. Deliverables

- **Jupyter Notebook (.ipynb)**: A documented workflow covering the code, requirements, and the model pipeline.

- **Technical Report**: A summary (2–4 pages) explaining the strategy, experimental changes, and an analysis of results.
- **DATASET LINK**: Xanadu00/autotrain-data-galaxy_classification · Datasets at Hugging Face

0.2 Problem Statement: The Apophis Deflection Mission

1. Background

The asteroid 99942 Apophis (approx. 340m diameter) is on a trajectory that will bring it dangerously close to Earth on April 13, 2029. While NASA has ruled out an impact for 2029, the asteroid will pass within 32,000 km of the surface—closer than our geostationary satellites. In this simulation, we are testing a "Kinetic Impactor" deflection strategy. We want to hit the asteroid with a spacecraft to slightly alter its velocity Δv, ensuring it misses Earth by a safe margin.

2. The Challenge

You must build a **3-Body Gravity Simulator** (Sun, Earth, Apophis) to predict the asteroid's path.
Your Goal: Find the optimal date to hit the asteroid and the minimum impulse (Δv) required to increase the 2029 Earth-flyby distance to at least **50,000 km** (from Earth's center).

3. Physics Constraints

- **System**: You must simulate the gravitational pull of the Sun and Earth on the asteroid (Moon gravity can be ignored for simplicity).
- **Physics Engine**: Use Newton's Law of Universal Gravitation.
- **Integrator**: You must implement a symplectic integrator (e.g., Velocity Verlet or Runge-Kutta 4). **Do not use Euler integration** as it is too inaccurate.
- **Impulse**: The impact is instantaneous. The direction of Δv can be any vector (prograde, retrograde, or normal).

4. Input Data (Initial Conditions)

- **Simulation Start Date (Epoch)**: January 1, 2028 (MJD 61772.0)
- **Asteroid (Apophis) Orbital Elements at Epoch**:
  - Semi-major axis (a): 1.379×10^{11} m
  - Eccentricity (e): 0.191
  - Inclination (i): 3.34°
  - Longitude of Ascending Node (Ω): 204.4°
  - Argument of Perihelion (ω): 126.4°
  - Mean Anomaly (M): 135.0°

5. Deliverables

- **Simulation & Notebook**: Provide the code for the simulation and the notebook used to derive the solution.
- **Additional Analysis**: Including detailed 2-3 page report with your explanations or visualizations that help understand your solution.

0.3 Problem Statement: The Sentinel Shield – Near-Earth Comet (NEC) Classification

1. Background: The Planetary Defense Mission

Our solar system is a cosmic shooting gallery. Among the millions of objects orbiting the Sun, **Near-Earth Objects (NEOs)** —specifically comets and asteroids—pose a unique challenge to Earth's safety. While most bypass our planet at safe distances, a select few are designated as **Potentially Hazardous Objects (PHOs)**.

According to NASA/JPL CNEOS standards, a "Hazardous" classification is typically triggered by two critical physical thresholds:

- **Proximity**: A Minimum Orbit Intersection Distance (MOID) of **0.05 AU** or less.
- **Size/Magnitude**: An **Absolute Magnitude** of **22.0 or brighter**, suggesting the object is large enough (approx. 140 meters) to cause significant regional damage upon impact.

Identifying these threats early using orbital mechanics is the first line of defense for our planet.

2. Objective

The goal of this task is to develop a supervised machine learning model to automate the **Hazardous Classification** of Near-Earth Comets and Asteroids. You must build a system that can accurately predict whether a celestial body is a "Potentially Hazardous" threat based on its orbital elements and physical properties.

3. Data Description: Orbital & Physical Metadata

Participants will work with a high-dimensional dataset containing orbital parameters from 1950 to 2025. Key features include:

- **Physical Properties**: Absolute Magnitude and Estimated Diameter.
- **Orbital Shape & Tilt**: Eccentricity, Inclination, Semi-Major Axis, and Perihelion/Aphelion distances.
- **Proximity Metrics**: Minimum Orbit Intersection Distance (MOID) and Miss Distance.
- **Dynamics**: Relative Velocity, Orbital Period, and the **Jupiter Tisserand Invariant** (used to distinguish between comet-like and asteroid-like orbits).
- **Reliability**: Orbit Uncertainty and Orbit Determination Date.

4. Participant Tasks

- **Data Exploration**: Investigate the relationship between Absolute Magnitude and the "Hazardous" label. Analyze if certain orbital families (e.g., those with high eccentricity) are more likely to be hazardous.
- **Preprocessing**: Handle high-cardinality features (like names/IDs) and address the extreme class imbalance, as hazardous objects are statistically rare.
- **Model Training**: Implement classification algorithms.
- **Evaluation**: Focus on **Recall** (minimizing False Negatives) and **AUC-ROC**, as missing a hazardous object is far more costly than a false alarm.

5. Deliverables

- **Jupyter Notebook (.ipynb)**: A well-commented, end-to-end pipeline from data cleaning to model deployment.
- **Technical Report**: A summary focusing on your strategy for handling class imbalance and an analysis of approach.

Dataset Access

DATASET: click here
Note: If prompted, click "Download anyway" to download the file: `nasa_neows_1950_2025.csv` (201M).

Achilles, the Tortoise, and a Schwarzschild Black Hole

In the fifth century BCE, Zeno of Elea posed a challenge to common sense that has refused to die. Drawing on the figure of Achilles—the swiftest of Homeric heroes, forged in war and song—Zeno imagined him racing a humble tortoise granted a small head start. Achilles, embodiment of speed and mortal excellence, should overtake the animal without effort; yet Zeno argued otherwise. Each time Achilles reaches the place where the tortoise once stood, the tortoise has already moved ahead, and the chase fragments into an endless sequence of ever-diminishing intervals. Motion, so obvious to the senses, dissolves under analysis into an infinite regress. Zeno's paradox was not a denial of experience but an assault on its foundations: if space and time are infinitely divisible, how can motion ever be completed? In posing this question, Zeno transformed a simple race into a meditation on infinity, continuity, and the limits of human understanding—one whose echoes still shape the way we think about time, motion, and reality itself.

You are an experimenter with a peculiar and unsettling power: you can go back in time, trap Achilles himself inside a fragile spaceship and send them on a one-way plunge into a Schwarzschild black hole. The tortoise: patient, immovable, a symbol of paradox sits at the event horizon $r = 2M$. You have chosen the experiment precisely to test a paradox: will your falling Achilles ever "catch" the tortoise in your coordinate time t?

Work in geometric units $G = c = 1$. In the equatorial plane ($\theta = \pi/2$) the Schwarzschild line element is

$$ds^2 \;=\; -\left(1 - \frac{2M}{r}\right) dt^2 \;+\; \left(1 - \frac{2M}{r}\right)^{-1} dr^2 \;+\; r^2 d\varphi^2. \tag{1}$$

Let τ be the Achilles' proper time and denote derivatives with respect to τ by overdots $\dot{} \equiv d/d\tau$. Starting from the spacetime interval you will:

(a) Derive the Lagrangian for the system(It's expression is given by Eqn. 2). Show t and φ are cyclic; derive canonical momenta and conserved quantities E and L from the Lagrangian

$$\mathcal{L} = \tfrac{1}{2}\, g_{\mu\nu}\dot{x}^\mu \dot{x}^\nu \tag{2}$$

Show the canonical momenta and their conserved quantities:

$$p_t \;=\; \frac{\partial \mathcal{L}}{\partial \dot{t}} \;=\; -\left(1 - \frac{2M}{r}\right)\dot{t}, \qquad\qquad E \equiv -p_t \;=\; \left(1 - \frac{2M}{r}\right)\dot{t}, \tag{3}$$

$$p_\varphi \;=\; \frac{\partial \mathcal{L}}{\partial \dot{\varphi}} \;=\; r^2\dot{\varphi}, \qquad\qquad L \equiv p_\varphi \;=\; r^2\dot{\varphi}. \tag{4}$$

Argue why, by spherical symmetry, motion can be taken in the equatorial plane $\theta = \pi/2$ without loss of generality.

(b) Using the normalization condition for timelike geodesics,

$$g_{\mu\nu}\dot{x}^\mu \dot{x}^\nu = -1, \tag{5}$$

and the expressions for E and L, derive the radial equation in the form

$$\dot{r}^2 + V^2(r) = E^2, \tag{6}$$

and show algebraically that

$$V^2(r) = \left(1 - \frac{2M}{r}\right)\left(1 + \frac{L^2}{r^2}\right). \tag{7}$$

(c) Specialize to $L = 0$(Your experiment). Starting from the radial equation, show that for your spaceship starting from rest at $r = R$ one has

$$E^2 = 1 - \frac{2M}{R}, \qquad \dot{r}^2 = 2M\left(\frac{1}{r} - \frac{1}{R}\right). \tag{8}$$

Solve the differential equation and derive the cycloidal parametric solution

$$r(\chi) = \frac{R}{2}(1 + \cos\chi), \qquad \tau(\chi) = \frac{R^{3/2}}{2\sqrt{2M}}\big(\chi + \sin\chi\big). \tag{9}$$

Use the parametric form to deduce the total proper time to fall from $r = R$ to $r = 0$ (set $\chi = \pi$) and give the final expression in closed form.

(d) Show that

$$\frac{dr}{dt} = \frac{\dot{r}}{\dot{t}} = \frac{(1 - 2M/r)\,\dot{r}}{E}, \tag{10}$$

directly from $E = (1 - 2M/r)\dot{t}$. Define the tortoise coordinate

$$r_* \equiv r + 2M \ln\left|\frac{r}{2M} - 1\right|. \tag{11}$$

Show algebraically that

$$\frac{dr_*}{dt} = \frac{\dot{r}}{E}. \tag{12}$$

Using the radial equation $\dot{r}^2 = E^2 - V^2(r)$, argue that as $r \to 2M^+$ (provided E is nonzero and large enough to surmount any potential barrier) we have $\dot{r} \approx -E$ and therefore $dr_*/dt \approx -1$. Integrate this asymptotic relation to obtain the late-time dependence

$$r(t) \approx 2M + C\,e^{-t/(2M)} \tag{13}$$

for some constant C. Be explicit about which variable is exponentiated (the Schwarzschild coordinate time t) and how to convert the geometric-units result into SI seconds when needed (restore factors of G and c as appropriate).

(e) Use the result of (d) to answer this: Do you ever see Achilles reach the tortoise($r = 2M$) in finite Schwarzschild time t? Justify your answer using the asymptotic formula and the behavior $r(t) \to 2M$ as $t \to \infty$. Explain how this resolves the Achilles–tortoise paradox: in what sense does Achilles "reach" the tortoise, and why the two descriptions (proper time τ vs coordinate time t) are both physically valid yet different.